Supplementary material

From routine cine CMR to regional myocardial dysfunction

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8 October 2026

Supplementary Methods: literature search, screening and evidence handling

Review type. The review is a critical narrative review with a structured database search. It was not conducted as a systematic or scoping review: there was no registered protocol, no duplicate screening and no meta-analysis, and no PRISMA flow is claimed. The records described here are provided so that the boundary of the search and the depth of reading behind each claim can be checked.

Core set and citation expansion. The corpus began from a core set of papers: DeepStrain and its evaluations [1–3], StrainNet [4], the multivendor comparison of feature tracking with tagging [5], the DENSE and tagging comparisons [6–8], the deformation-standardisation and SCMR consensus documents [9–12], and the reviews published from 2019 to 2026 [13–17]. Their reference lists and citing articles were followed, and guideline, dataset and repository records were added by hand because bibliographic databases index them incompletely. Expansion stopped when a further round changed no method family, comparison axis or principal conclusion. This is a thematic stopping rule; it does not establish exhaustive coverage.

Structured search. PubMed, Europe PMC and OpenAlex were queried through their official interfaces on 3 September 2026 with a publication cutoff of 1 September 2026. Six concept families were searched in each database, giving 18 query–database combinations: regional cine strain after infarction; known-motion or digital-phantom validation; perturbation of the prior or regularisation; conditional recoverability; failure-aware or selective prediction; and regional localisation, spillover and attenuation. The exact strings, hit counts and retrieval dates are in `supplementary/historical/gap-search-query-log.csv`. The three databases reported 41, 244 and 508 hits. Retrieval was capped at 100 records per query; the cap truncated the broad post-infarction query in Europe PMC (239 hits) and OpenAlex (473 hits) and affected no other combination. The export contained 281 rows and 253 unique records after merging by DOI, PMID or OpenAlex identifier and normalised title.

Screening and retention. The purpose of the structured search was to establish whether any study had combined a prescribed regional abnormality, known motion and a controlled change of the estimator, and to collect studies that address part of that design. Titles and available abstracts were screened once against that purpose, with software assistance for query execution, deduplication and screening suggestions; the retention decisions are the author's responsibility and were not repeated by a second reviewer. A record was retained when it contributed at least one element of the design: a prescribed or known abnormality, known motion, a controllable prior, a common estimand, a regional retention or spillover outcome, or failure-aware assessment. Twenty-one records were retained and 232 were set aside at the title-and-abstract stage; no full-text exclusion

stage was applied. A directed check on 9 September 2026, with the same cutoff, searched for cine-only tissue-inference studies and added two records absent from the 253 (Righetti and colleagues, 2025; Yang and colleagues, 2025, preprint). The retained set is therefore 23 records; two of them report MulViMotion (journal and preprint versions), so the set represents 22 studies. The design elements present in each record are listed in `supplementary/historical/near-neighbour-matrix.csv`. Within this boundary no retained study implemented all elements together, and the main text uses this result only in the bounded form stated in Sections 3.3 and 4.1 of the main text. Retained records entered the review in the same way as records from citation expansion: a record is cited only where its primary source was read at the depth stated in the evidence ledger.

Verification update. A targeted verification pass was completed on 8 October 2026. Exact-title and DOI searches were run for the studies that carry the principal quantitative statements: StrainNet; the multivendor tagging comparison; the DENSE, tagging and feature-tracking comparisons; the cine–DENSE comparison; Strain-8; the reconstruction studies (CineVN, REGAIN, DENT and model-based deep-learning reconstruction against compressed sensing); the recent finite-element and physics-informed estimators; Fourier-Net+; CardioMorphNet; and the 2026 single-slice DENSE/cine study. Official pages were used for the CMRxRecon2025 release and modality statements. The bibliographic metadata of every reference were checked against Crossref, arXiv or DataCite (`evidence/crossref_verification_20261009.csv`). Reviews published from 2024 to 2026 were read to position the contribution; no priority is claimed.

Reading depth. The evidence ledger records, for each claim and source, the locator and the depth of reading. "Full text" means that the relevant methods, results, tables, discussion and limitations were inspected. "Abstract" means that only the abstract and bibliographic record were inspected; such a source supports only publication status, study design and results stated in the abstract. "Official record" denotes a dataset, repository or guideline page. Of the 135 claim–source pairs in the ledger, 75 were checked at full text, 56 at abstract level and 4 against an official record. Entries in Supplementary Table S3 are abstract- or metadata-level readings of the cited method papers unless the main text states a deeper reading; they describe each study's own reported evaluation and are not a re-analysis. "Not inspected", "not reported" and "not validated" are kept distinct throughout.

Software assistance. OpenAI Codex and Claude Code (Anthropic) assisted with query execution, deduplication, screening suggestions, ledger construction, bibliographic checks and consistency checks, as declared in the main text. Neither tool acted as an independent reviewer, and no screening, extraction or appraisal decision is attributed to them.

Supplementary Note 1: strain definitions and comparison traps

Let $\boldsymbol{\varphi} : \Omega_0 \to \Omega_t$ map a reference position $\boldsymbol{X}$ to its deformed position $\boldsymbol{x}$. The displacement is

$$\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{\varphi}(\boldsymbol{X}, t) - \boldsymbol{X}, \tag{1}$$

and the deformation gradient is

$$\boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}} = \boldsymbol{I} + \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}}. \tag{2}$$

Common finite-strain measures include the right Cauchy–Green tensor $\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}$, Green–Lagrange strain $\boldsymbol{E} = (\boldsymbol{C} - \boldsymbol{I})/2$, and logarithmic strain obtained from a stretch tensor. In one dimension, engineering strain $e = (\ell - \ell_0)/\ell_0$ and Green–Lagrange strain satisfy $E = e + e^2/2$. Numerically comparing values without

converting the definitions can create an apparent method bias [7].

Cardiac components require a coordinate frame. Circumferential, radial and longitudinal axes can be defined at the reference state, updated through time or constructed from a ventricular centreline. Local axes become unstable near the apex and can vary with segmentation. A transmural average differs from endocardial strain because deformation varies through the wall. Two-dimensional short-axis analyses omit longitudinal and through-plane derivatives; a reported radial or circumferential component is consequently conditional on the 2D model.

Peak definitions also differ. If $E_s(t)$ is the segment-average strain curve, $\min_t E_s(t)$ is generally not equal to the spatial average of pointwise minima. "End-systolic", "peak systolic" and "peak over the whole cardiac cycle" may diverge in dyssynchrony or post-systolic shortening. A reproducible comparison must fix the order of differentiation, projection, spatial averaging and temporal peak selection. Supplementary Figure S1 illustrates these four traps.

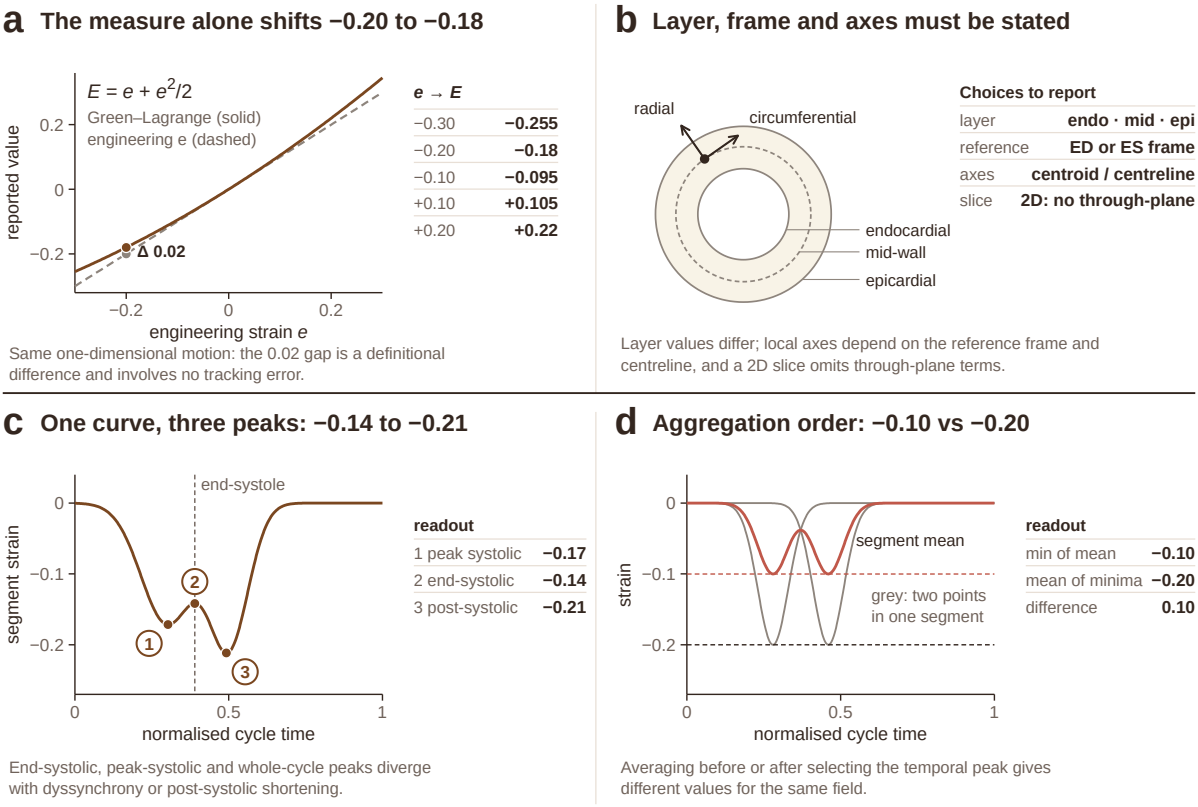


Figure S1. The same motion gives different numbers under different definitions. (a) Engineering strain e and Green–Lagrange strain $E = e + e^2/2$ differ by 0.02 at $e = -0.20$; the table converts five values. (b) Endocardial, mid-wall and epicardial layers and local radial and circumferential axes, with the choices a report must state (layer, reference frame, axes, 2D slice). (c) One segmental curve read as peak systolic (−0.17), end-systolic (−0.14) and post-systolic peak (−0.21). (d) The minimum of a segment-mean curve (−0.10) differs from the mean of pointwise minima (−0.20). Conceptual and analytic schematics; the printed values are computed from the drawn curves and have no measured scale. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

Supplementary Note 2: validation targets

The claim boundaries for common validation targets are summarised in Figure 5a of the main text.

A validation hierarchy should not be interpreted as a single ranking. Simulation supplies exact truth but imperfect realism. DENSE and tagging provide material-sensitive in-vivo information but have sequence- and processing-specific error. Scan–rescan establishes precision. LGE and outcomes test biological or clinical

relationships. A mature claim often needs several targets because their weaknesses are complementary.

When two acquisitions are compared, acquisition registration is part of the reference. Slice position, breath-hold, loading and heart-rate differences can create apparent algorithm error. Where reference processing includes smoothing or temporal fitting, its effective resolution should be reported. Segment-level analyses require participant-aware uncertainty estimates because segments from one heart are correlated.

Supplementary Note 3: error attributes and mapping validity

The distinct spatial, temporal and field-validity failure modes are illustrated in Supplementary Figure S2.

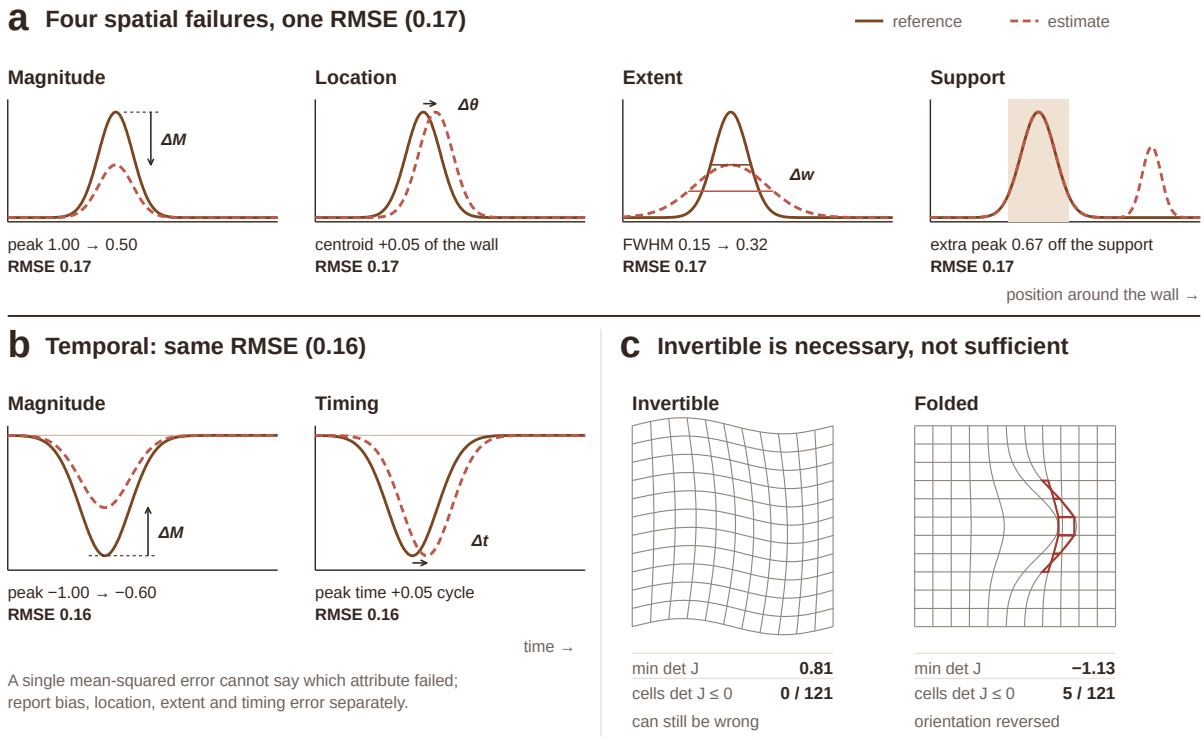


Figure S2. Regional error attributes and mapping validity. (a) Reference and estimate profiles separating magnitude, location, extent and support errors. (b) Temporal magnitude and timing errors. In a, and again in b, every estimate is set to the same root-mean-square error against its reference (0.17 and 0.16), so a single mean-squared error cannot tell these failures apart. (c) A smooth invertible mapping (minimum $\det J$ of 0.81), which can still be wrong, and a folded mapping with $\det J \leq 0$ in 5 of 121 outlined cells (minimum -1.13). Prescribed schematics with no measured scale. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

Let a prescribed abnormality field be $g_{\mathrm{ref}}(\boldsymbol{x}, t)$ and an estimate be $g_{\mathrm{est}}(\boldsymbol{x}, t)$. A single mean-squared error can conceal clinically different failures. Attribute-specific losses may include:

- magnitude error in an agreed region;
- distance between abnormality centroids or peaks;
- width, area or overlap error of a thresholded abnormal region;
- onset, peak-time or curve-alignment error; and
- error conditional on the reference support and a sensitivity analysis over the threshold.

Field validity is related but distinct. A positive Jacobian determinant is a useful check against local folding, while near-incompressibility may be physiologically motivated. Neither proves accurate correspondence.

A smooth, invertible mapping can be wrong, and a reference generated by an approximate 2D model can violate three-dimensional tissue behaviour. Validity metrics should accompany, not replace, regional accuracy metrics.

Supplementary Note 4: learning target and control point

DeepStrain and StrainNet illustrate why the target matters. DeepStrain demonstrated a complete automated workflow with segmentation, motion estimation, volumetric agreement, reference ranges and repeatability analyses [1]. Its cine motion network was not independently validated against dense material displacement for every regional claim. StrainNet used DENSE phase-derived displacement as a supervised target, but it reported two distinct evaluations [4]. In the 62-person independent test set, its 0.75 ± 0.35-mm pixelwise EPE was obtained when DENSE-magnitude contours, not cine contours, were supplied to the network and compared with DENSE displacement over myocardium and time. Separately, bSSFP cine feature-tracking contours were used for the end-systolic global and AHA-segmental circumferential-strain comparison in 59 test participants. Thus the EPE and cine-contour ICCs must not be fused into one "cine displacement" result; population, contour source, registration, direction, aggregation and time-point still bound the claim.

Training-time and inference-time controls must also be distinguished. Changing a loss weight usually requires matched retraining, and varying it only at test time has no effect unless the trained model exposes that parameter. Post-processing strength or a biomechanical solve, by contrast, can be a genuine inference-time control. Evaluation should compare like with like, hold splits and estimands fixed, and keep final test cases out of model or rule selection.

Supplementary Figure S3 shows the two control points and their shared held-out evaluation contract.

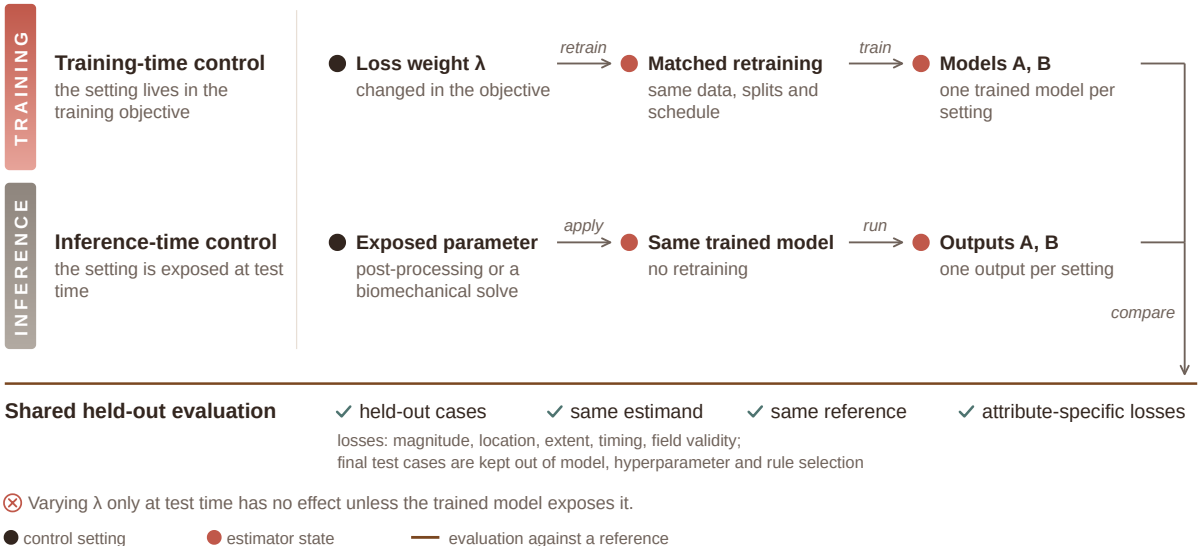


Figure S3. Training-time versus inference-time controls. A training-loss weight requires matched retraining; a test-time parameter can be varied only when the implementation exposes it. Candidate models or outputs are compared on the same held-out cases under the same estimand, reference and attribute-specific losses. Conceptual schematic. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

Supplementary Note 5: pre-specified evaluation template

For each intended-use claim, the following items can be completed before the final test set is analysed:

Item	Pre-specified content
Intended use	Clinical/research decision; user; consequence of false regional call
Population	Disease, age, rhythm, prior treatment, exclusions and recruitment setting
Acquisition	Scanner/vendor, field strength, sequence, views, voxel size, slice gap, temporal resolution, acceleration/reconstruction
Abnormality	Direction, magnitude, location, width/extent, transmurality and timing range
Estimator	Name/version, inputs, user interaction, representation and regularisation
Estimand	Reference, strain measure, axes, layer, dimensionality, region, temporal summary and aggregation
Reference	Simulation/phantom, DENSE, tagging, landmarks, tissue target or outcome; registration and uncertainty
Primary loss	Attribute-specific error at subject level
Tolerance	Clinically/technically justified threshold and sensitivity analysis
Splits	Participant-, centre- and time-aware separation; no final-test selection
Subgroups	Scanner, centre, image quality, pathology, wall thickness, abnormality scale and demographic groups
Failure policy	Exclusion, imputation, quality control or abstention; observable signals and coverage
Statistics	Bias/limits, confidence intervals, repeated-measures model, multiplicity and missingness

Supplementary Note 6: software, versioning and regulatory context

Feature-tracking and strain tools change over time. Regulatory clearance may cover a broad software application or specified outputs but does not establish material-motion accuracy for every regional analysis. Product webpages document current availability, not historical algorithms. Studies should therefore record software name, exact version, analysis module, contour convention, manual corrections and any export transformation. Reference intervals and serial thresholds should not be transferred across products without a bridging study [5, 6, 18].

The source package does not bundle historical FDA summaries or manufacturer webpages and therefore does not claim that such records were independently audited in this correction pass. Bibliographic citations and dated review notes support only the limited interpretation above; they are not a continuously current market survey. Authors should verify any product indication, clearance, version and jurisdiction against the live primary record before making a clinical recommendation.

Supplementary Note 7: reconstruction and dataset scope

Accelerated reconstruction can alter temporal sharpness, noise texture and edge appearance, all of which may affect downstream motion estimation even when conventional image-quality scores remain acceptable. Primary evidence is encouraging but endpoint-specific. In REGAIN, global feature-tracking strain from 49 participants showed near-zero mean differences between separately acquired accelerated and GRAPPA breath-hold cine, with non-zero limits of agreement [19]. CineVN's paired 18-volunteer comparison found the smallest strain bias among the tested accelerated reconstructions, but all accelerated protocols retained systematic peak radial/circumferential bias; the 46-patient demonstration had no conventional reference and

included segmentation-related slice exclusions [20]. In ten healthy volunteers, model-based DLR improved the precision of global circumferential strain at high reduction factors relative to compressed sensing, while bias and radial underestimation remained [21]. DENT improved image-phase interpolation and reader-rated temporal presentation across a large multicentre development set and prospective testing, but it did not evaluate myocardial strain or validate its deformation vectors as tissue motion [22]. These measured findings motivate joint reconstruction–estimator tests; they neither show nor imply that deep learning erases focal abnormalities.

A reconstruction study that agrees on volumes, global strain or image appearance does not automatically validate regional correspondence, lesion location, spatial extent or timing. Ideally, the reconstruction and motion estimator are evaluated jointly against a material-sensitive or prescribed-motion reference, with the fully sampled or conventional reconstruction treated as a comparator that carries its own error. Temporal blurring, sharpening, alias suppression and hallucination are candidate mechanisms, not established explanations for a regional result unless directly tested.

CMRxRecon2025 is useful for reconstruction robustness because the official site lists multiple centres, vendors, disease groups and a broad set of CMR modalities [23]. The official description does not provide material-motion ground truth, and it does not justify assuming that cine and LGE are paired for every participant. The 2024 challenge included facts and tasks that should not be transferred to 2025. Dataset labels, access, pairing, licensing and intended task must each be verified before use.

Supplementary Note 8: study-level interpretation examples

Study	Strongest direct support	Important positive evidence	What remains open
StrainNet [4]	DENSE-contour pixelwise displacement EPE; separately, cine-contour end-systolic global/segmental Ecc	Higher global and segmental Ecc agreement than conventional FT in 59 independent-test participants	EPE is not a cine-contour result; focal location/extent, other directions, external domains and reference uncertainty
Militaru et al.[5]	Product-specific global/regional agreement with tagging and LGE discrimination	Strong global LS/CS agreement for tested products	Regional/radial interchangeability and material truth
Kihlberg et al.[8]	Scar-discrimination performance in suspected CAD	DENSE/tagging outperformed FT for high-transmurality LGE segments	Pointwise motion error and other diseases/thresholds
Strain-8 [24]	Same-day precision by modality/direction/field strength	Several global measures had good or excellent repeatability	Accuracy, disease cohorts and longer-term reproducibility
MRXCAT2.0 [3]	Technical capacity/error under generated cases	Known motion paired with realistic-looking cine	Unmodelled anatomy, artefact, scanner and disease complexity
Jahromi et al.[25]	Phantom and healthy-volunteer component-specific comparison	Circumferential recovery was stronger than other directions in tested setting	Pathology, base/apex, 3D effects and clinical thresholds

Supplementary Table S1: empirical validation contracts

Table S1. Detailed measurement contracts for selected empirical studies. NR means not reported in the inspected source at the required level; it does not mean absent from the field.

Study and denominator	Population, input and analysis unit	Output, scale, layer, reference state and time	Reference, statistic and result	Directly supported inference
Wehner et al.[7]; 88 participants; 186 paired slices; torsion and dyssynchrony subset 38	Healthy and heterogeneous cardiac diagnoses from two centres; bSSFP plus matched DENSE; subject and slice	Transmural whole-slice base/mid/apex peak Green–Lagrange Ecc; four-chamber longitudinal strain; torsion and curve-delay dyssynchrony, ED reference	DENSE; mid Ecc bias -0.4%, CoV 11%; base bias -4.0%, CoV 18%; apex bias -2.4%, CoV 15%; dyssynchrony bias 26 ms, CoV 76%	Mid-slice peak Ecc agreement under matched definitions; poorer basal/apical magnitude and temporal synchrony agreement
Cao et al.[6]; 95 recruited, 87 analysed (77 cardiomyopathy, 10 controls)	Matched tagging, DENSE and bSSFP analysed by three FT products; subject and one matched mid slice/4-chamber	Peak Lagrangian Ecc and Err in one mid short-axis slice; Ell in four-chamber; systolic peak	HARP tagging; DENSE Ecc ICC 0.778, bias 0.155 percentage points, 95% limits -6.166 to 6.476; all methods had poor Err agreement	Relatively close Ecc but technique-dependent Err/Ell; no focal location, extent or dense-field endpoint
Militaru et al.[5]; 61 participants	18 healthy, 18 chronic MI, 15 DCM, 10 LV hypertrophy; bSSFP with three FT products, 3 T; subject and AHA segment	Global and 16-segment transmural peak-systolic Lagrangian LS, CS and RS; ED reference	HARP tagging; global ICC 0.92–0.94 LS, 0.88–0.91 CS; vendor range 0.10–0.81 for global RS; LGE discrimination tested in MI	Strong global LS/CS agreement for tested products; regional/radial values and scar discrimination are product dependent
Kihlberg et al.[8]; 116 patients	Suspected CAD; DENSE, tagging, bSSFP FT and LGE; subject and segment	Global/segmental circumferential and radial strain and torsion; peak systolic; exact reference convention not extracted	LGE $>$ 50% transmurality; segment AUC 0.87 DENSE, 0.83 tagging, 0.66 FT; sensitivity at 80% specificity 82%, 71%, 35%	Technique-specific discrimination of high-transmurality scar; not a dense displacement ranking
StrainNet [4]; 305 total, 243 development/670 sections, 62 independent test/190 sections; cine analysis 59	Patients, healthy adults and healthy children/adolescents; DENSE magnitude contours for displacement test, separately bSSFP cine FT contours; pixel, subject and AHA segment	Intramyocardial displacement over frames; separately end-systolic global/segmental circumferential Green–Lagrange strain, ED reference	DENSE phase-derived displacement; EPE 0.75 ± 0.35 mm with DENSE contours; cine Ecc ICC 0.87 global/0.75 segmental versus FT 0.72/0.48	Positive independent-test displacement performance under DENSE-contour input and bounded cine end-systolic Ecc agreement; the two tests are not interchangeable
Strain-8 [24]; 20 healthy volunteers	Same-day rescan using cine, tagging and DENSE at 1.5/3 T plus 2D/3D echo; subject and segment	Global and AHA-segmental frame-wise peak strain; modality-specific layer/definition	Repeat scan; CoV, ICC and Bland–Altman; global CoV $\leq$ 20% across protocols, segmental values frequently $>$ 20%, especially longitudinal/radial	Precision and minimum detectable change under controlled same-day conditions; no accuracy or disease-cohort claim
Jahromi et al.[25]; deforming phantom and 40 healthy volunteers	One-slice cine model versus one-/two-slice DENSE models; subject, slice and endocardial/epicardial layer	Slice-wise Ecc, Ell and Err through the normalised cycle; mid-ventricular; model-specific reference	Prescribed phantom plus DENSE-model comparison; reasonable cine agreement for endocardial Ecc, directional/layer differences elsewhere; numerical limits NR in inspected abstract	Endocardial Ecc equivalence in healthy mid-slice setting; not base/apex, disease or focal lesion validation

Supplementary Table S2: validation-resource details

Table S2. Validation resources, access state and reference conditions as checked on 8 October 2026. The phrase not file-verified means that no subject-level archive was downloaded or processed in this review.

Resource	Nature, scale and population	Modalities, coverage and pairing	Motion reference and evaluable scale	Access, processing requirement, uncertainty and domain boundary
STRAUS [26, 27]	Synthetic: 18 cases generated from 3 volunteer templates, each with healthy, LBBB, LCX, LAD, RCA and RCA2 states	For each case: 3D cine MR, three tagged-MR sequences and 3DUS over the cardiac cycle; 90 sequences/2700 volumes reported	Framewise reference meshes provide dense correspondence; the paper reports regional engineering strain	Public/no login reported; MHD/RAW images and VTK meshes are ready files but require matched coordinate/time/reference handling; imposed mechanics, three anatomies and simulated appearance bound transfer

Resource	Nature, scale and population	Modalities, coverage and pairing	Motion reference and evaluable scale	Access, processing requirement, uncertainty and domain boundary
Stanford Cardiac Performance MRI [28]	Human: laboratory page 51 healthy; DOI metadata 55 total (51 healthy, 4 aortic stenosis), acquired 2023–2025	3 T whole-LV bSSFP cine; DENSE at corresponding locations with xyz encoding; three mid-LV tagged slices; T1 and cDTI described	DENSE phase and tagging can support in-vivo displacement/strain comparisons after processing; whole-LV cine coverage does not imply whole-LV tagged reference	Stanford Digital Repository, ODbL 1.0; subject ZIP DICOMs; not file-verified here; requires reconstruction/phase unwrapping, segmentation, registration and estimand alignment; cohort total is not a usable-pair count
CMAC [29, 30]	Dynamic phantom and 15 healthy volunteers; challenge database	Cine SSFP, 3D tagged MRI and 3DUS; whole-cycle 3DTAG and cine with modality registration described	12 landmarks (four walls at three LV levels) tracked by two observers in 3DTAG; sparse point error, not dense field truth	Open access with attribution; median inter-observer variability 0.77 mm phantom and 0.84 mm volunteers; healthy/sparse/legacy acquisition domain
MRXCAT [31]	Synthetic generator based on XCAT; not a fixed participant cohort	Cine and first-pass perfusion with cardiac/respiratory motion, multiple coils, noise, trajectories and undersampling	Prescribed XCAT motion and fully sampled simulation support sampling/reconstruction and known-motion tests	MATLAB framework available for non-commercial research by request; XCAT access and self-generation required; signal, texture, partial-volume and anatomy assumptions limit realism
MRXCAT 2.0 [3, 32]	Synthetic generator, not a fixed cohort; healthy, infarct, DCM and HCM exemplars	bSSFP-like cine generated from statistical LV shape, biophysical function and texturised tissue labels	Known simulated function/strain enables voxel/mesh-scale capacity and estimator tests	Public code repository with example material; generation and dependencies required; generated cases are not independent patients and model/texture coverage bounds transfer
CMRxRecon 2025 [23]	Human multicentre, multivendor, multidisease reconstruction challenge/resource; exact motion-reference denominator not applicable	Cine plus mapping, contrast-weighted, black-blood, phase-contrast, LGE and perfusion described; pairing varies by task/release	No declared dense material-motion truth; can assess reconstruction robustness and downstream domain shift if endpoints are added	Challenge terms and released reconstruction/QC files; disease labels are not motion truth and cine–LGE pairing cannot be assumed for every participant

Supplementary Table S3: cine-CMR motion estimators by year

Table S3. Motion and strain estimators discussed in the main text, in approximate chronological order within each family. Every cell is drawn from the cited source; NR means the source does not state the item, and no value has been inferred from a related paper. Rows are not ranked and differ in status (journal, conference, workshop, preprint); the year/source column states which is which. A short validation entry usually means the method was evaluated for the claim its own authors made, not that it was tested and found wanting for this review's question.

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Conventional estimators and comparators					
Boundary feature tracking	Family [33]	Cine plus endo/epicardial contours; per-case tracking	Contour continuity and propagation; interior motion largely interpolated	Many clinical cohorts; inter-software/version agreement, scan–rescan, tagging/DENSE comparison, clinical association	Many disease cohorts; regional and radial agreement consistently weaker than global [5, 24, 34]
Free-form deformation (FFD)	1999; IEEE TMI [35]	Image pair; none	Affine plus B-spline lattice with smoothness cost; NMI data term	Contrast-enhanced breast MRI; comparison with rigid/affine registration	Not applicable (non-cardiac origin); transformation model reused by later cardiac work
Harmonic phase (HARP)	2000; IEEE TMI [36]	SPAMM-tagged MR (comparator, not cine); none	Phase of an isolated spectral peak	Real and simulated tagged images; small-motion displacement and 2D strain	Not applicable; material-sensitive comparator

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Early clinical feature tracking	2009–2013 [37–39]	Cine plus contours; vendor feature tracking, per case	Proprietary feature tracking	30 post-STEMI patients (scar association); 191 DMD patients and 42 controls versus HARP, mid-LV whole-slice Ecc; 145 healthy volunteers with tagging in 20 (abstract level)	Scar association and whole-slice/global agreement; no focal localisation endpoint
Groupwise nD+t B-splines	2011; Med Image Anal [40]	Whole dynamic sequence; none	Spatio-temporal B-spline, optional cyclic constraint, no fixed reference	Synthetic and 3D+t lung/cardiac CT; sub-voxel accuracy; LV volume curves within about 3% of manual	NR
Laplace-based contour mapping	2015; Int J CARS [41]	Cine contours; none	Laplace equation between successive contours	Synthetic data and in-vivo feasibility (abstract level)	NR
Variational data assimilation	2016; IEEE TMI [42]	Cine sequence; none	Contraction model coupled to image observations	Synthetic data and a clinical cohort (abstract level)	NR as a prescribed focal experiment
Gradient-based optical flow	2019; Med Image Anal [43]	Cine and tagged MR; none	Intensity and shape-feature invariance, iteratively optimised	Realistic simulations and preliminary clinical data (abstract level)	NR
Automated segmentation plus spline tracking	2017; FIMH [44]; 2018; ISBI [45]	Cine; supervised segmentation, unsupervised tracking	Learned segmentation feeding a conventional tracker	42 HCM patients and 21 controls; pixel accuracy about 97% against manual segmentation	Group separation HCM versus controls; no motion reference, no focal experiment
Anatomy-Aware Tracker	2020; MLMI [46]	Long-axis cine; weak shape supervision	Convolutional VAE of realistic shapes refining a dense tracker	Tracking versus other methods; qualitative shape realism	NR
Groupwise locally low-rank	2024; BMC Med Imaging [47]	Whole cine sequence; none	Groupwise registration minimising patchwise signal rank	Simulation ($n = 20$), public cine ($n = 100$), in-house tagging patients ($n = 16$); point, voxelwise and global strain error; late-diastolic drift; tagging radial-strain correlation	Paired tagging in patients; disease composition NR; no prescribed focal abnormality
Unsupervised learning-based registration					
VoxelMorph	2019; IEEE TMI [48]	Image pair; unsupervised (optional auxiliary segmentations)	Amortised similarity-plus-smoothness objective	General registration datasets; accuracy and run time versus selected comparators	NR
TransMorph	2022; Med Image Anal [49]	Volumetric pair; unsupervised	Hybrid Transformer–CNN; diffeomorphic and Bayesian variants	Brain MRI (inter-patient, atlas) and phantom-to-CT; topology preservation; uncertainty calibration	NR for cardiac pathology
Fourier-Net	2023; AAAI [50]	Image pair; unsupervised	Band-limited Fourier representation of displacement; model-driven decoder	Registration benchmarks; accuracy with speed, memory and multiply-adds	NR
Fourier-Net+	2026; IEEE TNNLS [51]	Band-limited image representation; unsupervised	As Fourier-Net with shortened contracting path; cascaded variant	Five datasets including one 3D cardiac MRI set; Dice, Hausdorff distance, ED volume and EF versus TransMorph and other baselines at lower memory and compute (abstract level)	NR; cardiac evaluation is geometric/functional, not regional strain
Cardiac-specific estimators					
Joint motion and segmentation	2018; MICCAI [52]	Cine sequence; unsupervised motion, weakly supervised segmentation	Shared multi-scale encoder; Siamese recurrent spatial transformer	220 subjects; accuracy and speed versus competing methods	NR

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Apparent flow	2019; Med Image Anal [53]	Two cine phases; semi-supervised	Pixel-wise flow combined with masks	ACDC; classification accuracy 95% (train) and 94% (test) from nine features	Pathology classes are the target; motion validated via the downstream label
Motion Pyramid Networks	2020; MICCAI [54]	Cine pair; unsupervised, cyclic teacher–student	Multi-scale fused pyramid	Two public clinical datasets; multiple metrics and inference time	NR
FOAL	2020; CVPR [55]	Cine sequence at test time; meta-learned online optimiser	Test-time adaptation of a learned tracker	Two public datasets; accuracy under train–test mismatch; about 0.4 s per video	Distribution shift is the target, not characterised as focal abnormality
MulViMotion	2022; IEEE TMI [56]	SAX plus 2-/4-chamber LAX cine; unsupervised with weak shape supervision	Hybrid 2D/3D network; shape regularisation module	580 UK Biobank subjects; quantitative and qualitative comparison for 3D LV motion	NR
DeepMesh	2024; IEEE TMI [57]	SAX and LAX cine; unsupervised via differentiable rasteriser	Template 3D endo/epicardial mesh propagated to subject space	UK Biobank; vertex-wise 3D displacement with maintained correspondence	NR
Bidirectional registration network	2023; Comput Biol Med [58]	Cine sequence; unsupervised	Bidirectional reference-to-frame prediction with sequence context	Public cine data (abstract level)	NR
Low-rank groupwise deformations	2024; preprint (MSc thesis) [59]	Image sequence; unsupervised	Groupwise diffeomorphic registration towards a low-rank stack	NR; comparison of resulting low-rankness with other groupwise approaches	NR
MCM (Mamba)	2025; MICCAI workshop [60]	Cardiac-cycle image sequence; unsupervised	Bidirectional Mamba block; adjacent-frame motion decoder	Two public datasets; comparison with conventional and learned trackers	NR
SegMorph	2025; IEEE TMI [61]	Cine sequence; joint motion and segmentation	Recurrent variational representation	Public cine data (abstract level)	NR
CardioMorphNet	2026; Med Image Anal [62]	SAX cine volumes; recursive registration of segmentation maps without an intensity loss	Recurrent VAE with separate segmentation and motion posteriors; uncertainty map	UK Biobank and M&Ms; warped versus reference masks; uncertainty versus other probabilistic methods	M&Ms is multi-centre/multi-pathology, but the comparison is mask agreement, not focal motion accuracy
Bidirectional latent neural ODE	2026; Comput Med Imaging Graph [63]	Cine sequence; unsupervised	Continuous-time latent ODE; Lagrangian consistency	ACDC and M&Ms; temporal consistency and plausibility versus selected comparators	NR
STCMT-Net	2026; Med Image Anal [64]	4D CT and cine MRI; unsupervised	Unsupervised keypoints as anchors; dense field from matched-keypoint basis vectors	4D CT and cine datasets; comparison with competing methods; strain analysis separating cardiac conditions	Group separation by condition; no prescribed focal abnormality, no motion reference
Semi-supervised motion flow	2025; Comput Biol Med [65]	Full cine sequence; semi-supervised with ED distance-map weighting	Memory network over all frames	In-house 271 patients; EPE 1.02 mm versus 1.19 mm (retrained RAFT); global peak RS/CS correlation 0.83/0.90 (abstract level)	NR; no material reference or focal abnormality
IG-BiReg	2026; Phys Med Biol (online first) [66]	Cine pairs; unsupervised	Implicit-guided bidirectional residual registration	Anatomical registration improvement (abstract level)	NR
CMR foundation models	2025–2026 [67, 68]	36M images/27,524 subjects (self-distillation); 15M cine images/74,916 studies	Pre-trained representation fine-tuned per task	Segmentation, landmark localisation, diagnosis, prognostication across independent datasets	Dense motion estimation is not among the reported tasks

Physics-, biomechanics- and representation-constrained estimators

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Biomechanics-informed prior	2023; Med Image Anal [69]	Cine sequence; unsupervised, prior learned from simulated deformations	VAE manifold of plausible deformations traversed at tracking time	Three public cine datasets; tracking accuracy, volume preservation, generalisability, derived strain	NR as a prescribed focal experiment
WarpPINN	2023; Med Image Anal [70]	Cine bSSFP; per-case optimisation	Neural deformation with hyperelastic regulariser and Jacobian near-incompressibility; Fourier features	Synthetic examples; 15-volunteer cine benchmark; landmark tracking; strain by automatic differentiation	Healthy benchmark; focal abnormality not evaluated
WarpPINN-fibers	2026; Eng Comput [71]	Cine MR; per-case optimisation	As WarpPINN plus fibre-stretch penalty along synthetic fibres	Synthetic phantom and the same 15-participant benchmark; fibre versus fibre-free formulation	Healthy benchmark; focal abnormality not evaluated
Implicit neural representation	2026; STACOM [72]	NR	Implicit neural representation of intramyocardial motion and strain	NR (no abstract in the record retrieved)	NR
Space–time finite-element correlation	2026; Med Image Anal [73]	Routine SAX and LAX cine; correlation framework	Region-specific biomechanical regularisation; data-driven temporal decomposition; multi-view alignment	One synthetic dataset with ground-truth motion/strain; three public datasets with landmarks or masks; one clinical dataset with masks; versus two FT and four deep-learning baselines	One synthetic ground-truth setting; in-vivo references are landmarks and masks; abnormality size/severity not varied
Reference-supervised and end-to-end strain					
Cine-only infarct delineation	2018; Med Image Anal [74]	Non-contrast cine; supervised on manual LGE delineation	LSTM motion features plus optical flow, fused per pixel	165 cine datasets; pixel accuracy 95.0%, Dice 89.9%, $\kappa = 0.91$ against two radiologists	Infarction is the target; reference is LGE tissue, not motion
Survival-supervised motion	2019; Nat Mach Intell [75]	Cine-derived motion; supervised on survival	Learned low-dimensional motion representation	302 patients; Harrell C 0.75 (95% CI 0.70–0.79) versus human benchmark 0.59 (0.53–0.65)	Prognostic discrimination; neither regional accuracy nor focal abnormality evaluated
DeepTag	2021; CVPR [76]	Tagged MR (comparator acquisition); unsupervised	Bidirectional diffeomorphic registration; differentiable composition to Lagrangian field	Clinical tagging dataset; landmark tracking accuracy and efficiency	NR
3D tagging networks (synthetic training)	2025; JCMR [77]	3D tagged MR (comparator acquisition); trained on synthetic images from a biophysical LV model	Two networks predicting LV motion through the cycle	Synthetic 3D tagged images with analytic signal model (abstract level)	Not applicable; tagging analysis, not cine
DeepStrain	2021; Front Cardiovasc Med [1]; 2022 comparison [2]	Cine bSSFP; image, anatomical-label and smoothness losses	Segmentation plus motion network; end-diastolic reference; Green–Lagrange strain	ACDC ($n = 150$); CMAC landmark EPE 2.9 ± 1.5 mm; scan–rescan in ten volunteers; 47 healthy/533 patients versus manual FT, r 0.83–0.85 radial and 0.88–0.91 circumferential	Independent MRXCAT2.0 evaluation on normal, dilated, hypertrophic and infarcted cases (main text, known-motion tests)
StrainNet	2023; Radiol Cardiothorac Imaging [4]	Contour time series (DENSE-magnitude at displacement test; cine FT at strain test); supervised on DENSE displacement	2D+t CNN; prior implicit in DENSE-derived training distribution	243 development/62 independent-test participants (670/190 sections); EPE 0.75 ± 0.35 mm with DENSE-magnitude contours; cine Ecc ICC 0.87 global/0.75 segmental versus FT 0.72/0.48 in 59	Mixed patient population with material-sensitive reference; abnormality size/severity not varied; no focal location/extent endpoint
DL synthetic strain	2023; Radiol Cardiothorac Imaging [78]	Cine; supervised on radiologist wall-motion reference	NR in detail	53 patients/846 segments; AUC 0.90 against four-reader consensus at 30% peak radial strain threshold	Regional wall-motion abnormality is the target; reference is expert reading, not measured motion

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
DENSE-guided rotation network	2025; IEEE ISBI [79]	Routine bSSFP cine; supervised on DENSE displacement	Separate rotation and radial branches fused	NR in the record retrieved; strain accuracy against DENSE with large rotation as target	NR
DL aligned strain	2025; Radiol Artif Intell [80]	Cine; supervised, LGE-defined fibrosis reference	NR in detail	139 Duchenne patients; fibrosis evaluation in 57, reproducibility in 82; specificity +40%, accuracy +17% versus comparator	Target is LGE-defined fibrosis, a tissue reference
MyoNet	2026; Bioengineering [81]	Cine (rat model); supervised against sine-wave-modelling reference	CNN with alternating kernel sizes	Dahl salt-sensitive rats after radiation; ICC 0.973 circumferential, 0.975 radial	Rodent radiation injury; not human cine; analysis-derived reference

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